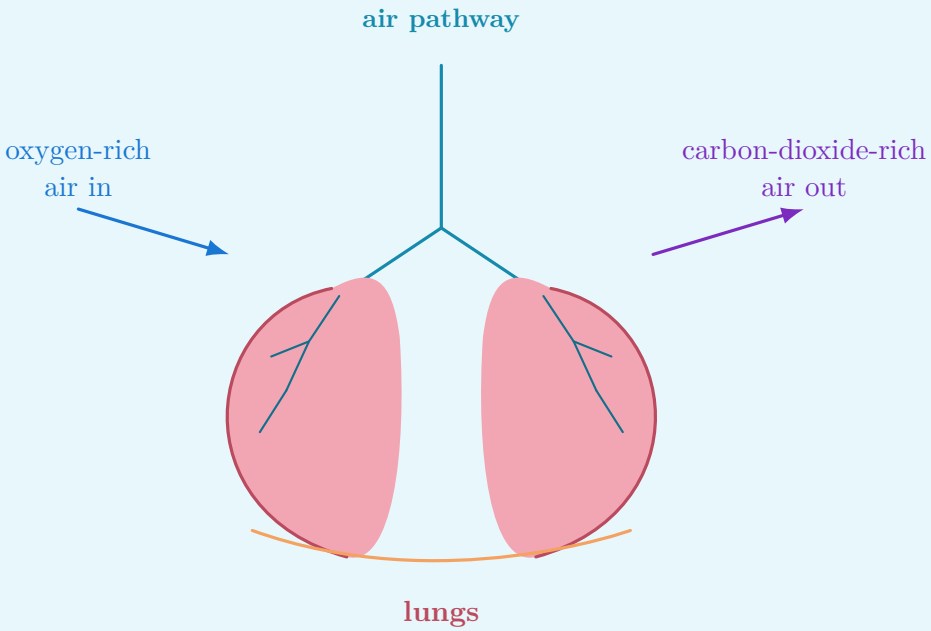
RESPIRATORY SYSTEM

& Respiration in Animals

Class 7 Science — Life Processes in Animals



Based on the uploaded Grade 7 *Curiosity* textbook, Chapter 9.
Main focus: **human respiratory system, breathing, gas exchange and respiration**.
Includes **all 10 end-of-chapter exercise answers first**; digestion-only questions are clearly marked as chapter-completeness items.

Contents

1 1. Textbook Exercise — Complete Answers 2

2 2. Breathing and Respiration Are Not the Same 3

3 3. Human Respiratory System: Parts and Air Pathway 4

3.1 Simple respiratory-system sketch 5

4 4. How Breathing Happens 5

4.1 Inhalation and exhalation 5

4.2 Activity 9.2: Bottle-and-balloon model 5

5 5. What Do We Breathe Out? 6

5.1 Activity 9.3: Lime-water evidence 6

5.2 Inhaled and exhaled air 6

6 6. Gas Exchange in the Alveoli 6

7 7. Respiration Releases Energy 7

8 8. Do All Animals Breathe the Same Way? 7

9 9. Health and Everyday Connections 7

10 10. In-Chapter Questions and Activity Answers 8

11 11. Exploratory Projects — Student-Ready Guidance 8

12 12. One-Page Concept Map 9

13 13. Common Mistakes to Avoid 9

14 14. Rapid Revision Test 9

1. Textbook Exercise — Complete Answers

Note: Questions 1, 2 and 9 are digestion questions. They are included here only so this Chapter 9 exercise set is complete; the main notes remain respiratory-system focused.

Q1. Complete the journey of food through the alimentary canal

Food → Mouth → Oesophagus (food pipe) → Stomach → Small intestine → Large intestine → Rectum → Anus.

Q2. Chapati / chewed chapati / potato + iodine

- **A: pieces of chapati** — turn **blue-black** because starch is present.
- **B: chewed chapati** — may show **no blue-black colour or only a faint blue-black colour** because saliva breaks down much of the starch into simpler sugars.
- **C: boiled mashed potato** — turns **blue-black** because potato contains starch.

Q3. Role of the diaphragm in breathing

Correct option: (iii) To help in inhalation and exhalation.
During inhalation it moves downward; during exhalation it moves upward.

Q4. Match the respiratory parts with their functions

Part	Correct function
Nostrils	(a) Fresh air from outside enters
Nasal passages	(d) Tiny hair and mucus trap dust and dirt
Windpipe	(e) Air reaches our lungs through this part
Alveoli	(b) Exchange of gases occurs
Rib cage	(c) Protects lungs

Q5. How can Sanvi show that breathing and respiration are different?

She can ask questions such as:

- "Does breathing only move air in and out, while respiration releases energy from glucose?"
- "Is breathing a physical process but respiration a chemical process?"
- "Does respiration use oxygen to break down glucose and produce carbon dioxide, water and energy?"

These questions show that the two processes are connected but **not the same**.

Q6. What do we inhale?

Tanu is the most scientifically complete: "We inhale air rich in oxygen."
Air is a mixture of gases. Inhaled air contains about **21% oxygen**; it is not pure oxygen.

Q7. Why do we sneeze after inhaling dust-laden air?

Dust particles can irritate the nasal passages. Hair and mucus trap many particles, and sneezing helps force irritating material out of the nose. Thus, sneezing acts as a protective response.

Q8. Why might Anusha breathe faster than Paridhi after running?

Possible explanations include:

1. Anusha may have run **faster or with greater effort**, so her body needed more oxygen and produced more carbon dioxide.
2. The two students may differ in **fitness, stamina or recovery rate**; Anusha may need more breaths per minute to meet the body's demand after exercise.

Core idea: stronger activity generally increases the body's demand for oxygen and removal of carbon dioxide.

Q9. Rice flour experiment with saliva

Yadu wants to test the **action of saliva on starch**.
Test tube A (without saliva) should show a stronger blue-black iodine colour because starch remains.
Test tube B (with saliva and enough time) should show little or no blue-black colour because saliva breaks down starch into simpler sugars.

Q10. Lime water with inhaled/surrounding air vs exhaled air

Rakshita is investigating whether **exhaled air contains more carbon dioxide than the surrounding air we inhale**.
She can confirm this by comparing the lime water:

- Test tube A, with surrounding air, shows little/no cloudiness.
- Test tube B, with exhaled air, turns **milky/cloudy more clearly**.

Since carbon dioxide turns lime water milky, this supports her conclusion.

2. Breathing and Respiration Are Not the Same

Key Idea

Breathing is the physical process of **inhaling** air into the lungs and **exhaling** air out of the lungs. **Respiration** is a chemical process inside the body in which oxygen helps break down glucose to release energy.

Feature	Breathing	Respiration
Type	Physical process	Chemical process
Main role	Brings in oxygen-rich air and removes air with more carbon dioxide	Releases usable energy from glucose
Main place discussed	Respiratory system, especially lungs	Inside the body/cells
Word equation	—	Glucose + Oxygen → Carbon dioxide + Water + Energy

Memory Trick

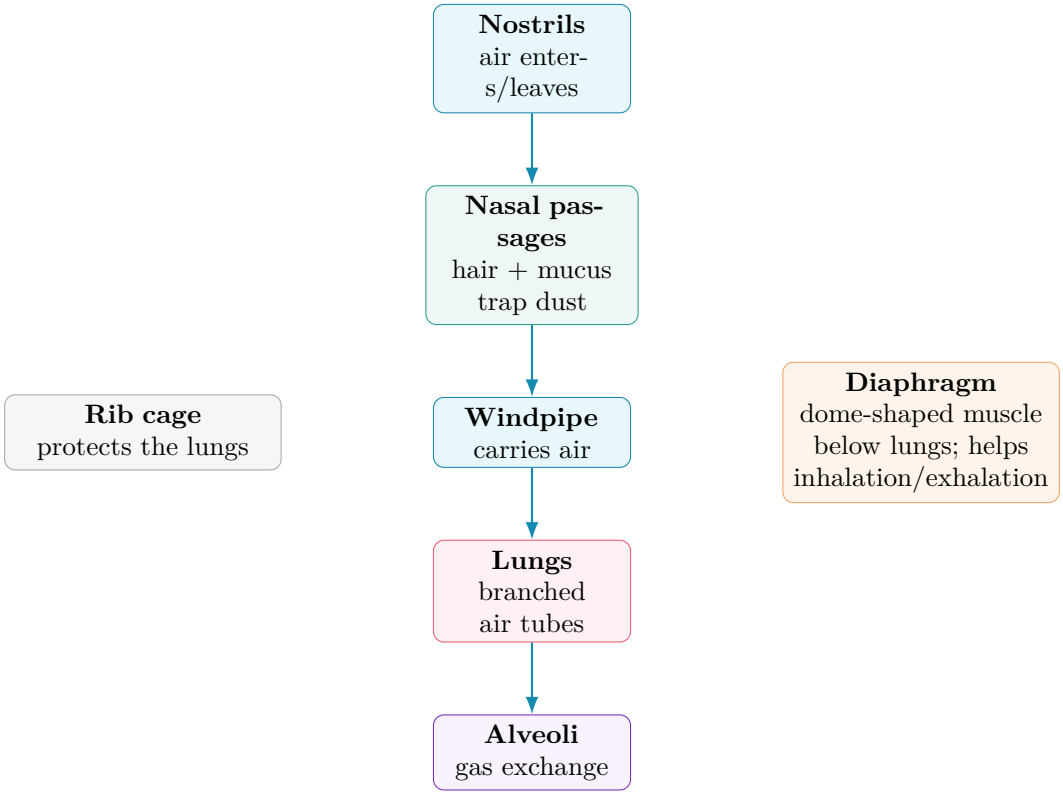
Breathe = move air. Respire = release energy. Breathing supplies oxygen; respiration *uses* oxygen.

Common Mistake

Do not write "we inhale oxygen". We inhale **air rich in oxygen**. Air also contains other gases.

3. Human Respiratory System: Parts and Air Pathway

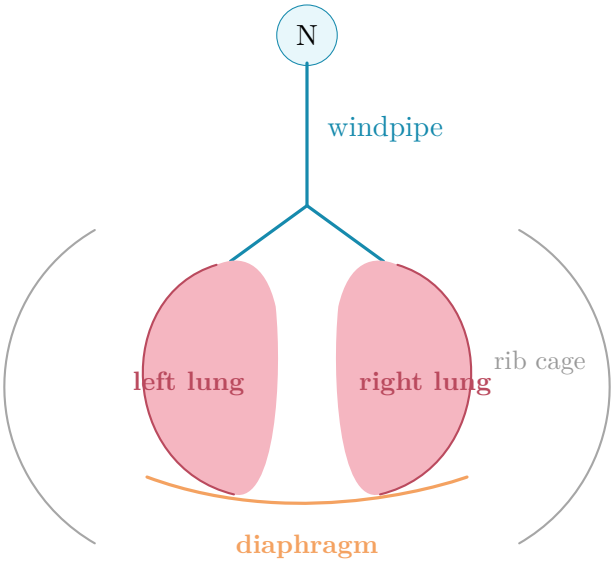
The respiratory system provides a pathway for inhaled and exhaled air and enables exchange of oxygen and carbon dioxide.



Why is nose-breathing preferred?

Tiny hair and mucus in the nasal passages help trap dust and dirt before the air reaches the lungs.

Simple respiratory-system sketch



4. How Breathing Happens

Inhalation and exhalation

	Inhalation (breathe in)	Exhalation (breathe out)
Ribs	Move up and outwards	Move down and inwards
Diaphragm	Moves downward	Moves upward
Chest space	Increases	Decreases
Air	Enters lungs	Is pushed out



Activity 9.2: Bottle-and-balloon model

Activity / Evidence

In the textbook model:

- the **two balloons represent the lungs**;
- the **rubber sheet represents the diaphragm**;
- pulling the rubber sheet **down** makes the balloons inflate;
- letting the rubber sheet move **up** makes the balloons deflate.

This models how diaphragm movement changes the space in the chest and helps move air.

Memory Trick

Down = Draw air in. Up = air goes out.
Diaphragm down → more chest space → inhalation.

5. What Do We Breathe Out?

Activity 9.3: Lime-water evidence

The textbook compares surrounding air with exhaled air using lime water.

Test tube A	**Test tube B**
surrounding air passed through lime water	exhaled air blown through lime water
little/no visible cloudiness	**turns milky/cloudy**

Key Idea

Lime water turns milky in the presence of carbon dioxide. Therefore, the activity shows that **exhaled air contains more carbon dioxide than inhaled/surrounding air**.

Inhaled and exhaled air

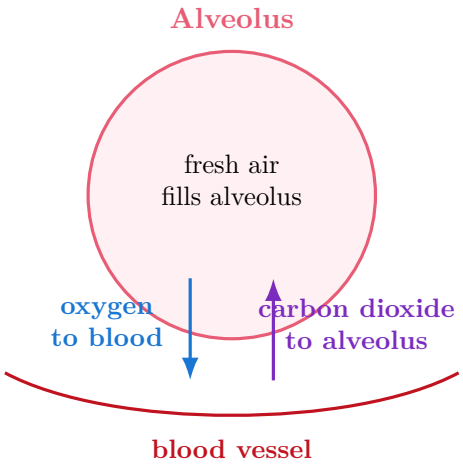
Gas	Inhaled air	Exhaled air
Oxygen	Nearly 21%	Nearly 16–17%
Carbon dioxide	Nearly 0.04%	Nearly 4–5%

Common Mistake

Exhaled air is **not oxygen-free**. It still contains about 16–17% oxygen according to the textbook figure.

6. Gas Exchange in the Alveoli

Alveoli are tiny balloon-like air sacs at the ends of the fine branches inside the lungs. They have thin walls and are surrounded by fine blood vessels.



Key Idea

Oxygen moves from the alveoli into the blood. **Carbon dioxide** moves from the blood into the alveoli and is later exhaled.

7. Respiration Releases Energy

The oxygen carried by blood is used to help break down glucose and release energy.

Glucose + Oxygen ⟶ Carbon dioxide + Water + Energy

This energy helps us walk, run, play and even think. The circulatory system helps transport oxygen and nutrients to body parts and carries waste products away.

Memory Trick

Air pathway gets oxygen to the lungs. **Blood pathway** carries oxygen from the lungs to the rest of the body.

8. Do All Animals Breathe the Same Way?

Different animals use different breathing structures suited to their habitats.

Animal / group	Main breathing structure described in the textbook
Humans, birds, elephants, lions, cows, goats, lizards, snakes	Lungs
Most aquatic animals such as fish	Gills
Tadpoles	Gills
Adult frogs	Lungs on land; skin for gas exchange in water
Earthworms	Moist skin

Adaptation idea

The textbook conclusion is that different animals have different breathing mechanisms adapted to their habitats.

9. Health and Everyday Connections

Health Link

The textbook notes that smoking damages the lungs and increases the risk of serious respiratory illness. Smoke can also harm non-smokers through passive smoking.

Breathing practices

The chapter mentions practices such as pranayama, Tummo breathing and deep breathing in the context of respiratory health, relaxation and concentration.

Why do we sneeze in dusty air?

Dust irritates the lining of the nasal passages. Sneezing is a protective response that helps expel irritating particles; nasal hair and mucus also help trap dust.

10. In-Chapter Questions and Activity Answers

What do the balloons and rubber sheet in Fig. 9.9 represent?

Balloons = **lungs**; rubber sheet = **diaphragm**.

What happens when the rubber sheet is pulled down?

The balloons inflate. This models inhalation: diaphragm moves down, chest space increases and air enters the lungs.

What happens when the rubber sheet moves upward?

The balloons deflate. This models exhalation: chest space decreases and air is pushed out.

Why does exhaled air turn lime water milky faster?

Because exhaled air contains **more carbon dioxide** than inhaled/surrounding air.

How does gas exchange happen?

Oxygen passes from the alveoli into the blood, while carbon dioxide passes from the blood into the alveoli to be exhaled.

11. Exploratory Projects — Student-Ready Guidance

The textbook gives open-ended projects. The notes below separate what is directly supported by the chapter from useful extension guidance.

Project: Air quality / AQI and respiratory health

Chapter link: inhaled air reaches the lungs; dust and small particles can affect the respiratory system.
Student task: compare how workers such as farmers, factory workers and street vendors may be exposed to dust, smoke or polluted air. Record likely sources of exposure and possible protective practices.
For actual AQI values or health effects, use a current trusted source; the textbook does not provide AQI ranges.

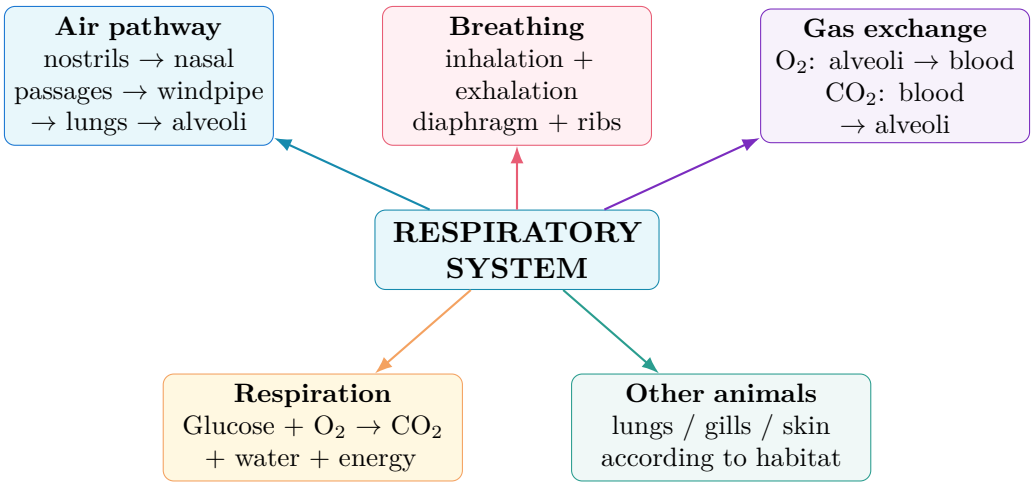
Project: Box breathing

Textbook prompt: read about box breathing and its benefits.
A simple pattern often used for learning is: breathe in, hold, breathe out, hold for equal counts. This is an **extension beyond the textbook's detailed content**; use a teacher-approved source for benefits and safe practice.

Project: Why can birds fly at high altitudes?

The textbook asks students to investigate how bird respiratory systems may be adapted for low-oxygen conditions. This is a **research question**; the chapter itself does not provide the detailed mechanism. Use a trusted biology source and clearly label findings as extension research.

12. One-Page Concept Map



Memory Trick

5-step story: Air enters → reaches alveoli → oxygen enters blood → oxygen helps release energy from glucose → carbon dioxide returns to lungs and is exhaled.

13. Common Mistakes to Avoid

Common Mistake

- **Wrong:** Breathing and respiration are identical. **Correct:** breathing is physical; respiration is chemical.
- **Wrong:** We inhale pure oxygen. **Correct:** we inhale air rich in oxygen.
- **Wrong:** Exhaled air contains no oxygen. **Correct:** it still has about 16–17% oxygen.
- **Wrong:** Gas exchange happens in the windpipe. **Correct:** gas exchange occurs in the alveoli.
- **Wrong:** Diaphragm moves up during inhalation. **Correct:** it moves down during inhalation.
- **Wrong:** All animals breathe only through lungs. **Correct:** fish use gills; earthworms use moist skin; frogs use different structures at different stages/conditions.

14. Rapid Revision Test

1. What is breathing? *Inhaling and exhaling air.*
2. Name the pair of openings through which air enters. *Nostrils.*
3. What traps dust in the nasal passages? *Tiny hair and mucus.*
4. Which tube carries air toward the lungs? *Windpipe.*
5. Where does gas exchange occur? *Alveoli.*
6. What happens to the diaphragm during inhalation? *It moves downward.*
7. Which gas is higher in exhaled air than inhaled air? *Carbon dioxide.*
8. Write the respiration equation. *Glucose + Oxygen → Carbon dioxide + Water + Energy.*
9. Is breathing physical or chemical? *Physical.*
10. Is respiration physical or chemical? *Chemical.*
11. How do fish breathe? *Through gills.*
12. How do earthworms exchange gases? *Through moist skin.*